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T28 – Case Study: Using receptor density-informed CMC priors for modelling mismatch negativity

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Receptor-informed Priors for DCM of Auditory MEG Responses

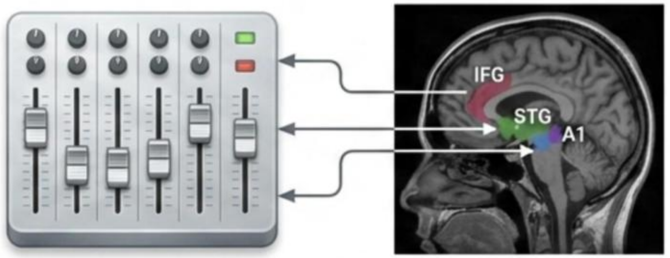
Can receptor-informed CMC intrinsic-connectivity estimates derived from iEEG spectra be incorporated as priors to a time-domain DCM of auditory MEG responses, and increase model evidence relative to standard prior expectations?

- 20 Healthy Controls
- Standard and Pitch Deviant Auditory Responses
- Baseline Model versus Receptor-Informed Model

Baseline Model: Standard Priors

Assumes uniform distribution of neurotransmitter receptor densities across cortical regions

Biologically-Informed Model: Receptor-Informed Priors

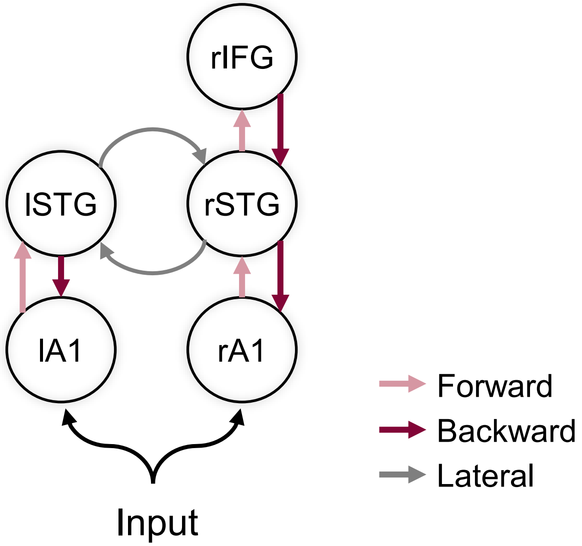


Incorporates regional variation in neurotransmitter receptor densities across cortical regions

RESEARCH QUESTION

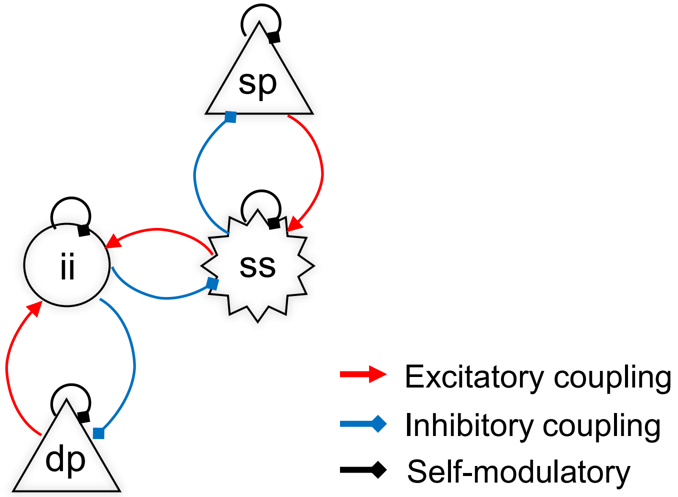
Transferring iEEG-derived Estimates to MEG Source Priors

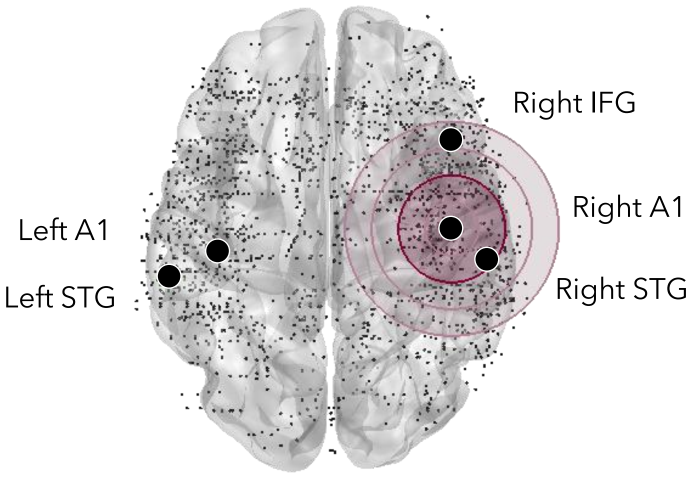
Five-Source Network Model



Each cortical source is represented by the a single Canonical Microcircuit (CMC)

Canonical Microcircuit Model





1,770 iEEG contacts, 5 Sources

Spatial Weighting of iEEG Estimates to MEG DCM Priors

iEEG CMC priors
10 G parameter priors (population coupling) at each iEEG contact

→

Euclidean distance (contact to source)
distance from ROI vs. iEEG coordinates

→

Gaussian kernel
convert distance to weight; normalise weights

→

Source-specific priors
kernel weighted mean of G priors (5 × 10 prior matrix)

DCM Specification: Different Prior Expectations on Intrinsic G Parameters

The baseline and receptor-informed DCMs fitted the same MEG data with the same network architecture. The model comparison isolated the effect of changing the prior expectations on intrinsic G parameters.

Constant across models

- Standard and Pitch Deviant responses
- Five-source MMN network
- Same CMC at each source
- Same Data Features
- Same non-G priors and covariance terms

Difference Between Models:

Prior Expectations of Intrinsic G Parameters

- Intrinsic G prior expectations
- 50 parameters = 5 sources × 10 G parameters

Baseline model

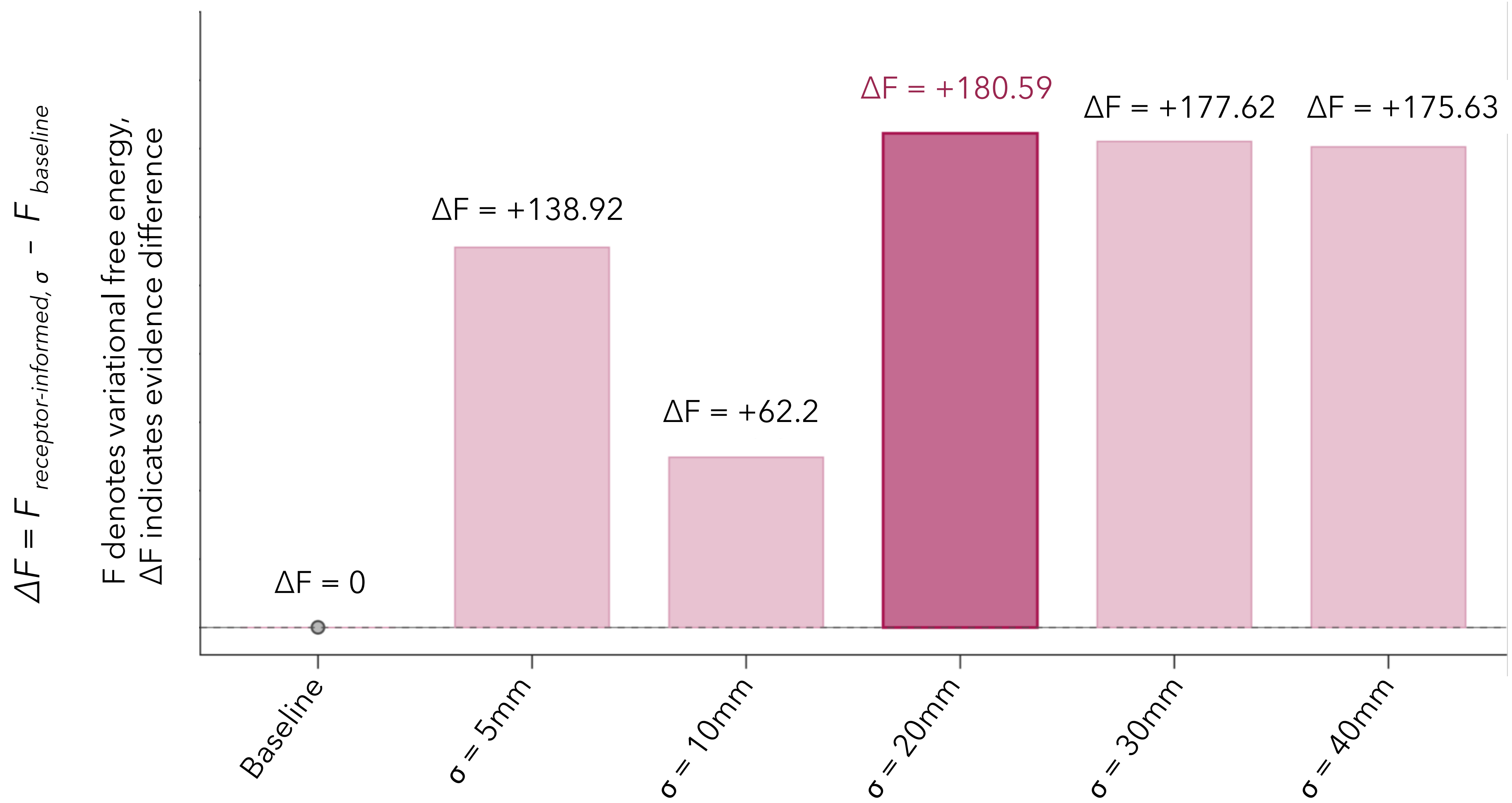
G prior expectation = 0
for every intrinsic connection

Receptor-informed models

G prior expectations = source-specific 5 × 10 matrix from spatially weighted iEEG estimates

Interpretation: any free-energy difference is attributable to the prior specification, because the data, architecture and inversion settings were matched.

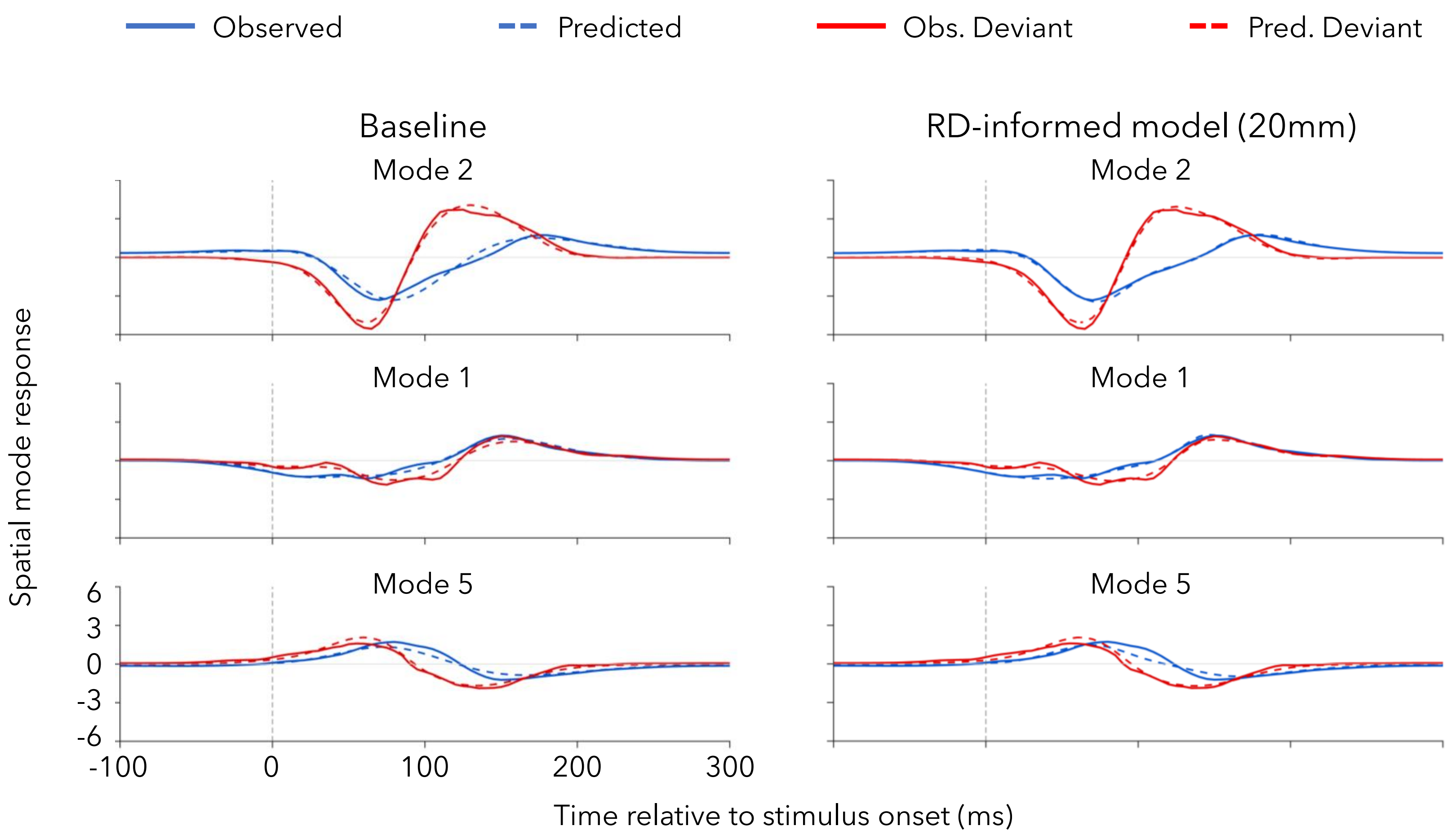
Receptor-informed Priors Increased Evidence Across Spatial Kernels (σ)



All tested receptor-informed models had positive ΔF relative to the baseline.
The most significant difference occurred at σ = 20 mm (ΔF = +180.59).

Assessing Fits of Baseline and Receptor-Informed Models (σ = 20 mm)

Fit of the Baseline and Receptor-Informed Model (σ = 20 mm)



Fit across displayed modes

- DCMs fitted Standard and Pitch Deviant responses jointly
- All eight spatial modes were included in inversion
- Displayed modes account for 82.9% of pooled variance

Baseline Model

R^2 = 0.972

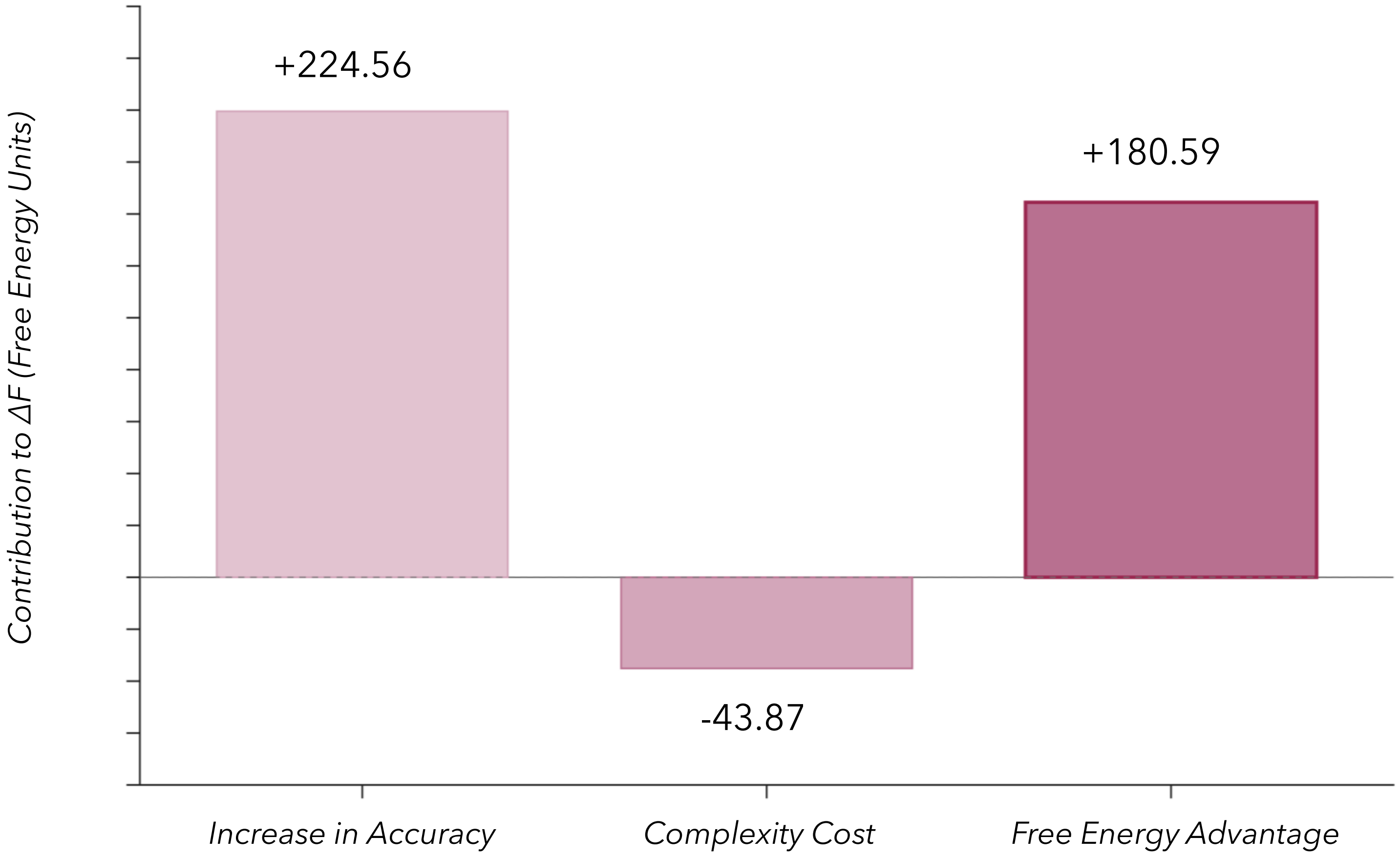
RMSE = 0.201

RD-informed Model (σ = 20 mm)

R^2 = 0.981

RMSE = 0.168

Decomposing the Evidence of the RD-informed Model (σ = 20 mm)



For the 20 mm RD-informed model:

- Accuracy increased by +224.46
- Complexity also increased by +43.87
- Net free energy advantage: ΔF = +180.59

Key Takeaway

Source-specific intrinsic G prior expectations derived from spatially weighted receptor-informed iEEG CMC estimates produced higher variational free energy relative to the baseline model with canonical prior expectations.

Spatially Shuffled Null Model: Assessing the Importance of Cortical Location

Key Question:

- Does the gain in free energy depend on assigning the iEEG-derived G estimates to the correct cortical locations?
- Or is it mainly driven by the empirical parameter values, regardless of where they are placed?
- **Test:** keep the same empirical G values and the same DCM but disrupt their anatomical assignment.

Empirical spatial assignment

True atlas
G estimates at original MNI locations

Weight to MEG sources
same Gaussian kernel and source locations

Empirical ΔF
evidence for anatomically assigned priors

Spatial-shuffle control

Shuffle locations
permute G vectors across cortical coordinates

Recompute priors
same values, same model, broken spatial mapping

Null ΔF distribution
repeat shuffles to obtain a null ΔF distribution

Interpretation:
if empirical ΔF exceeds the shuffled-null distribution, the spatial organisation of the priors contributes beyond the empirical parameter values alone.

Question: Does the RD-informed advantage depend on preserving the spatial correspondence between iEEG-derived CMC G estimates and their cortical locations, rather than on the empirical distribution of G estimates alone?

Open Questions

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